



Kenya's Steel Industry in a Globalized Economy

Steel Industry Paper
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Table of Contents

CHAIRMAN'S MESSAGE	4
EXECUTIVE SUMMARY	5
1. Introduction.....	7
2. Metal industry in Kenya	9
2.1. Historical background	9
2.2. Structure of the metal industry in Kenya.....	9
2.3. Facts and figures about steel industry in Kenya	11
2.4. Challenges facing the steel industry in Kenya	12
2.5. Steel Industry before and after imposition of enhanced import duties.....	13
3. Global and regional trends in the steel industry and associated challenges.....	14
3.1. Facts and figures of the global steel industry.....	14
3.2. Challenges facing the global steel industry.....	16
3.3. Impact of COVID-19 on the global steel industry	18
4. International experiences meant to support domestic steel industry	20
5. Steel production, sustainable development and emerging issues	26
5.1. Steel production in the circular economy	26
5.2. Climate change mitigation	29
5.3. Safety and health	31
5.4. Water management in the steel industry	31
6. Domestic opportunities that can be harnessed to promote the Steel Sector in Kenya	32
6.1. Developing domestic capacity to supply raw materials	32
6.2. Inter-industry linkages with the Steel Sector.....	33
7. Conclusion and policy implications	36
7.1. Conclusion	36
7.2. Policy implications	37
References.....	38
Annex 1: Steel industry value chain	40

List of Tables

Table 1.1: Economic benefits of steel industry in Kenya	8
Table 2.1: Export of steel products (MT)	11
Table 2.2: Import of steel products (MT)	12
Table 3.1: Ten leading steel producing countries in the world (million tonnes)	15
Table 3.2: Exports of semi-finished and finished steel products (MT, million) for some selected countries ..	16
Table 3.3: Imports of semi-finished and finished steel products (MT, million) for some selected counties ..	16
Table 4.1: Products under 14 categories for base and special grades	23
Table 5.1: Some examples of common uses of steel industry co-products	28
Table 5.3: Global projects to reduce CO ₂ production in the steel manufacturing process	29
Table 5.4: Water types in iron and steel industry	31

List of Figures

Figure 1.1: Leading producers of steel in the world in 2019.....	7
Figure 3.1: Average growth rate of crude steel production per annum (%)	14
Figure 3.2: Global crude steel production (million tonnes)	15
Figure 3.3: Evolution of global steel demand, 1950-2016.....	17
Figure 4.1: China pricing steel production below own cost of production.....	20
Figure 5.1: Benefits of steel in a circular economy.....	26
Figure 5.2: Indexed global energy consumption/ton of crude steel production	28
Figure 5.3: Life cycle of steel	29
Figure 5.4: Four-stage efficiency review process	30
Figure 6.1: Value of scrap metal exports in Kenya (Ksh. million).....	33
Figure 6.1: Number of locally assembled and imported vehicles in Kenya (number).....	34

CHAIRMAN'S MESSAGE

Steel is a core piece of the modern economy. It is used as a paradigm to gauge the overall economic status of a country, as well as an indicator of the investments dedicated to infrastructure and subsequently, a measure of developmental progress and stability.

Kenya's steel sector dates back to the pre-independence period, with the first firm established in 1948. The sector has since grown from the manufacture of nails to include wire drawing, galvanizing, hot-rolling and re-melting, cold-rolling, galvanized sheeting, and pipes.

The players within the sector have also grown in numbers and scope, and consequently, supported the country's socio-economic development goals through job and wealth creation. This has been realized through the manufacture of high-quality products for both local and export markets.

Ten years back, the steel sector was on its deathbed with many companies shutting down and turning their facilities into real estate. This trend has since changed following incremental government support to the sector in recent years, which led to the revival of the industry.

Despite this, the sector is heavily dependent on imported raw materials because Kenya's local deposits of iron ore and coal are yet to attract commercial interest. This calls for the country to unearth the sector's potential by incentivizing local industries to harvest raw materials from local deposits of iron ore and coal. As this entails huge capital investments, we call upon the government to accord investors with all the support needed to boost their competitiveness including banning exports of natural resources without being value-added (iron ore). This will not only encourage local industries to make this a reality but also lead to the creation of new jobs and wealth for many in the community.

The direct and indirect consumption of steel in the country is projected to increase as the country embarks on the development activities as envisioned in Vision 2030. The sector, therefore, has an opportunity to enhance the country's infrastructure through the purchase of locally manufactured steel products, further, bolster the growth of players within the value chain.

As we move towards this, it is critical that the sector adopts sustainable manufacturing practices. This includes increasing the focus on decarbonization through investments in energy efficiency, electrification and increased use of scrap metal and efforts to discover low carbon emission steelmaking technologies. Another critical factor is increasing the focus on the life cycle of and circular economy characteristics of steel. Sustainable partnerships are also key. For instance, the sector can partner with players along the value chain to incorporate Environmental, social, and governance (ESG) in their business operations and better management of steelmaking materials involving sorting and beneficiation processes to reduce environmental footprint.

The future of the steel industry is promising. This paper provides critical policy recommendations to fully realize the full potential of the sector to spur economic growth.

The Association remains committed to engaging our partners and stakeholders, to resolve sector challenges, by implementing the recommendations made herein. We look forward to partnering with to realize this agenda and our industrialization goals.

Bobby Johnson
KAM METAL AND ALLIED SECTOR CHAIR



... unearth the potential of the sector by incentivizing local industries to harvest raw materials from local deposits of iron ore and coal. This however requires huge capital investments

EXECUTIVE SUMMARY

The Steel industry is the foundation of any country's infrastructural development, from transport to housing.

The Steel sector is integrated with other economic sectors, such as the defence, transport, heavy engineering, energy and construction. China controls over 50% of the global steel industry. 10 steel-producing countries control almost the entire global value chain, accounting for about 84% of the world's steel output in 2019.

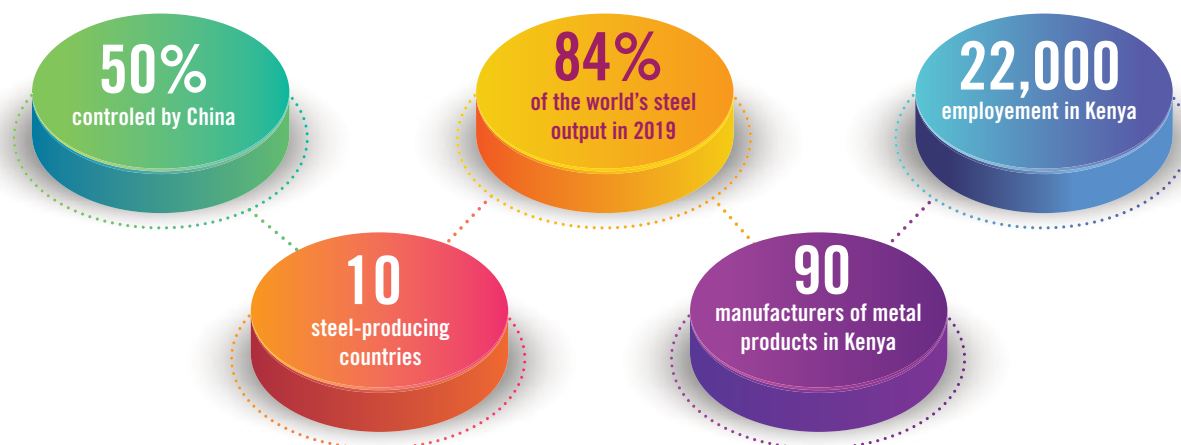
There are 90 manufacturers of metal products in Kenya, directly employing over 22,000. Kenya's Metal Sector comprises six sub-sectors, which include smelting/hot rolling/foundry and forgers, cold rolling/ galvanizing/colour coating, wire products and allied industries, pipes and tubes, general fabricators and aluminium industries.

Unfortunately, there are several factors that hinder the sector from realizing its full potential. Some of the main challenges are global overcapacity particularly by China, high cost of electricity, high dependence on imported raw materials, levies such as 1.5% for both Import Declaration Fee (IDF) and Railway Development Levy (RDL) and the importation of undervalued and substandard goods.

Global overcapacity caused by China's overproduction of steel threatened to collapse the industry in many countries and reduced their respective capacity for utilization. As a result, countries such as Brazil, USA, India and Columbia instituted specific measures to address competition from China. Some of these measures include increasing import duties and, in some instances, specific duties on some selected steel products.



Locally, the government imposed at least 25% and a specific duty rate of USD 200 on some selected steel imports to address undervaluation, misdeclaration and unfair competition from imports in 2016.



Locally, the government imposed at least 25% and a specific duty rate of USD 200 on some selected steel imports to address undervaluation, misdeclaration and unfair competition from imports in 2016.

This gave the steel industry a lifeline and attracted huge investments in the sector. Consumers have also benefitted from affordable, good quality domestic products as a result of increased competition among local manufacturers. However, increased duties shall not remedy increased competition from Egypt. Thus, the competitiveness challenge must be addressed. Competition is also set to increase once trading under the African Continental Free Trade Area (AfCFTA) commences.

As the world embraces circular economy, with consumers becoming more aware on matters sustainability, it is important that the steel industry in Kenya keeps up with emerging trends. As such, it needs to remain cognizant of global best practices related to steel production in the circular economy through recycling, use of co-products, energy efficiency, use of finished steel and life cycle approach to production. Others include climate change mitigation, safety and health considerations and water management. The need to embrace these emerging issues has been further amplified by the COVID-19 pandemic, due to the emphasis on green transition and the role of technological advances.

To further enable the steel industry in Kenya to play its rightful role in supporting the attainment of Kenya's development goals, some critical interventions are needed. These include granting protection sought during the annual budget making process against unfair trade practices, increasing the export levy for scrap metal from 20% to 35%, full operationalization of the Scrap Metal Council, creating a steel fund, exploiting available coal and iron deposits and creating a preferential electricity tariff rate for steel producers to reduce the cost of production.

With the country undertaking major infrastructural projects in transport and housing, it is now more important than ever, that Government creates policies that will drive the sector's growth. Additionally, promote consumption of local content in these initiatives, which shall lead us to prosperity, and position Kenya as a manufacturing powerhouse.

Phyllis Wakiaga
KAM CHIEF EXECUTIVE



1. Introduction

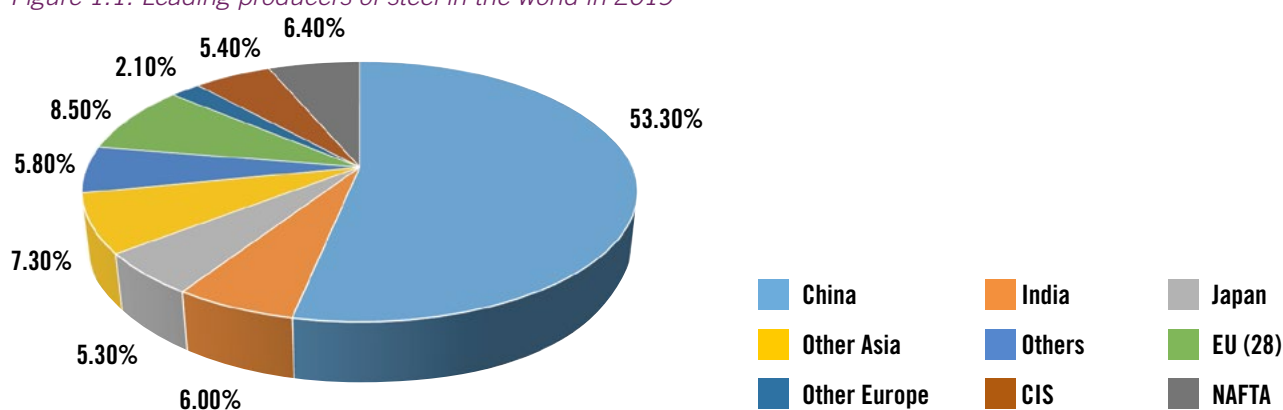
Steel is the world's most important engineering and construction material.¹ It is a widely held view that the iron and steel industry is a basis of development for a number of industries in a given economy, such as the defense industry, transportation, heavy engineering, energy and construction (including aeronautical and shipping construction) (Deloitte, 2017). In 2017, the global steel industry sold about US\$ 2.5 trillion worth of products, created US\$ 500 billion value added and directly employed about 6.1 million people.² According to the World Steel Association (Worldsteel), for every two direct jobs created in the steel sector, 13 others are created throughout the value chain.³

Steel is the backbone of the economic activity of any country. The performance of the steel industry is used to gauge the health of an economy. This is because the sector tends to be cyclical with increasing demand during economic booms and decreasing during recession.⁴ Per capita steel consumption is an internationally recognized indicator of the level of development of any country.

Direct and indirect consumption of steel in Kenya is projected to increase as the country embarks on development activities, as envisioned in Vision 2030 and the Big Four Agenda. The major Vision 2030 projects include the development of the Lamu Port, , railway and roads projects, housing, industrial parks and the development of special economic zones (SEZs), all of which utilize steel products. The iron and steel industry in Kenya accounts for about 13% of the manufacturing sector and in turn contributes significantly to the gross domestic product (GDP).

At the global scene, China is the leading producer of steel in the world and accounted for about 53.3% of global production in 2019, the EU about 8.5% while the rest of Asia approximately 7.3% (Figure 1.1). It is estimated that, in 2019, Africa accounted for only 0.9% of global steel output.⁵

Figure 1.1: Leading producers of steel in the world in 2019



Source: World Steel Association, 2020a

1 <https://www.worldsteel.org/about-steel.html>.

2 <https://www.worldsteel.org/steel-by-topic/statistics/steel-industry-economic-impact-.html>.

3 https://www.worldsteel.org/en/dam/jcr:7a759646-a2c6-4f7c-9c54-9710c0eb33e2/Full_steel_impact_infographic.pdf.

4 <https://ww2.frost.com/frost-perspectives/challenges-and-opportunities-steel-industry/>.

5 <https://www.worldsteel.org/en/dam/jcr:f7982217-cfde-4fdc-8ba0-795ed807f513/World%2520Steel%2520in%2520Figures%25202020i.pdf>.

There are two sets of steel industry value chains:

- a) **Integrated steel producers:** Steelmakers process the raw material as well as manufacturing output.. Plants have furnaces for blast burning of raw materials, rolling, casting and shaping of the ore into different uses.⁶ Integrated mills handle everything in one unit-processing raw materials, steel making, rolling and shaping. There are no integrated steel producers in Kenya because mining of iron ore and coal is yet to commence. Annex 1 gives a pictorial illustration of an integrated steel mill.
- b) **Secondary producers/mini steel plants:** These are small-sized manufacturers producing steel from sponge iron and scrap.⁷

Based on estimates by the Metal Sector, quantified economic contributions of the various sub sectors in the steel industry in Kenya are highlighted in Table 1.1.

Table 1.1: Economic benefits of steel industry in Kenya

Sub Sector	No	Direct employment	Indirect Employment	Installed Capacity Tons /annum	Utilized Capacity Tons /annum	Power Consumption (MW) per annum	Tax contribution (Ksh. billion) per annum	Value (Ksh. b) per annum
Smelting, Hot Rolling, Foundry & Forgers	27	10,000	30,000	2,268,000	630,000	254	10	47
Cold Rolling, Galvanizing & Colour Coating	4	2,000	2,000	505,000	325,000	10	9	40
Wire Products & Allied Industries	17	2,300	2000	300,000	150,000	7	3.5	15
Pipes & Tubes	19	2,000	20,000	980,000	323,000	14	6	27
General Fabricators	22	5,000	20,000	150,000	60,000	2	4	9
Aluminium Industries	1	800	1000	28,000	12,000	60	1	5
Total	90	22,100	75,000	4,231,000	1,500,000	347	33.5	143

Source: Metal Sector estimates

6 https://www.google.com/search?source=hp&ei=As5fXsHaDZGpgAbZgLEw&q=integrated+steel+plant&oq=integrated+steel&gs_l=psy-ab.1.0.0l10.376.4101..7964...0.0..0.1086.6418.2-4j8j1j0j1j1.....0....1..gws-wiz.....0i131j0i70i249.Xm5s4LweYyl

7 https://rbsa.in/archives_of_research_reports/RBSA_Indian_Steel_Industry_Analysis.pdf.

2. Metal industry in Kenya

2.1. Historical background

The metal industry in Kenya dates back to the pre-independence period. The first firm was established in 1948.⁸ Initially, the industry started by producing mainly nails and over time, integrated backwards into wire drawing, galvanizing, hot-rolling and re-melting, cold-rolling, galvanized sheeting and manufacture of pipes.⁹

Kenya United Steel Limited (KUSCO) was the first steel plant in Kenya, having started to manufacture wire products (nails) in 1949 and the first steel rolling and smelting plant in 1967.¹⁰ There is paucity of literature on the metal industry in Kenya. Nonetheless, the sector has grown in terms of numbers of operators and diversity.

2.2. Structure of the metal industry in Kenya

The Metal Sector, as constituted under KAM, comprises of six sub-sectors as described below.

a) Smelting/hot rolling/foundry and forgers

Steel smelting involves the conversion of steel scrap into an intermediary raw material known as 'ingots' or 'billets' (depending on the process) which are then used for Hot Rolling into finished steel products. The primary input for the process is steel scrap and ferro-alloys. There are over 21 melting furnaces in Kenya, with an installed capacity of approximately 600,000 tons per annum for producing ingots and billets. Kenya is estimated to produce about 288,000 tons of crude steel in electric induction furnaces annually. The sector is a major consumer of electricity with consumption estimated at around 187 MW annually.

Hot rolling involves the process of converting the intermediary raw material input, 'ingots' or 'billets,' through pre-heating the feed stock and then rolling, through a rolling mill, to various desired finished products. Examples include reinforcement bars, angle irons, channels, flat bars, window sections, light to medium sections among others. The intermediary stock of ingots/billets is available locally but most of the billets are imported. There are over 24 Rolling Mills in Kenya with an installed capacity of approximately 2.2 Million tons per annum for producing reinforcing bars (rebars), angles, and flats and sections. These products are essentially used in the construction of buildings, roads, bridges, ships and other fabricated items such as gates and road culverts.



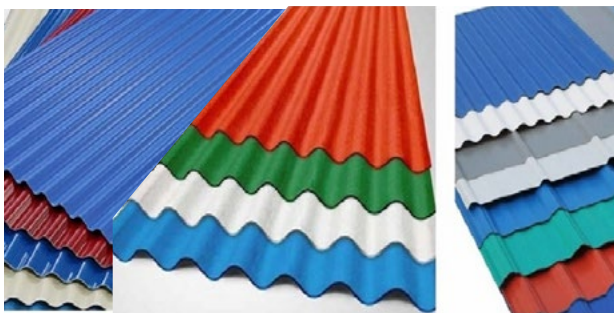
⁸ <http://archive.unu.edu/unupress/unupbooks/uu34ee/uu34ee0s.htm>.

⁹ <http://archive.unu.edu/unupress/unupbooks/uu34ee/uu34ee0s.htm>.

¹⁰ <http://archive.unu.edu/unupress/unupbooks/uu34ee/uu34ee0s.htm>.

Foundry engineering is the process of making castings in molds by melting metal and pouring it into molds to form metal objects. Castings obtain their shapes principally when molten metal solidifies in the designated forms. Ferrous metals (steel and cast iron), copper base, non-ferrous metals (brass, bronze etc.) and aluminium alloys can be cast in a foundry to get the required shapes. Currently, there are over 20 foundries in the country with a total installed capacity of 10,000 tons per annum.

b) Cold rolling/galvanizing/color coating



Three cold rolling mills are already in operation in Kenya. Inputs to these mills are hot rolled coils, pickled and oiled of varying widths based on the end use. Four continuous metal coating Manufactures exist with an installed capacity of 505,000 metric tonnes per annum. Three color coating companies engage in sheet color coating in the country. They have an installed capacity of around 100,000 metric tons per annum. The current utilized capacity in this sub sector is approximately 325,000 tons per annum.

c) Wire products and allied industries



The wire and wire products sector imports raw materials, which include hot rolled wires and rods, in various sizes and grades. The raw material is then processed for black and galvanized products such as nails, weld mesh, binding wire, reinforcing bars, rivets, barbed

wire, welding rods, bolts and nuts. Zinc which is used for galvanizing and lead for annealing are imported. Statistics held by the KNBS show that companies active in this sector have an average installed capacity of 300,000 tonnes per annum. The current scenario in Kenya is that we have over 17 manufacturers of wire related products operating at an average capacity of 150,000 tons per annum. The sector has been a huge attraction to investors. Blue Nile Rolling Mills LTD is in the process of completing a US\$ 19 million galvanizing wire plant in Thika with an annual installed capacity of 30,000 MT. Another investor, Roofings Kenya LTD, is setting up a US\$ 25 million at the Coast to produce galvanized wire and is set to be commissioned in early 2021.

d) Pipes and tubes



The manufacture of pipes and tubes involves the processing of black and galvanizing pipes of ½" to 10" inches and hollow sections of ½" to 8" inches. Products from this activity are used mainly by engineering and construction companies, the motor industry, fabrication and electrical works. According to the Ministry of Industrialization, Trade and Industrialization, the total installed capacity is about 980,000 tons per annum, whilst utilization standing at 323,000 tons per annum.

e) General fabricators



Industries involved in fabrication are heavy engineering industries, manufacturers of machines including tea and coffee machinery, minibus (*Matatu*) and bus body builders, trailer manufacturers, boiler makers as well as manufacturers of electrical switchgear and fittings. Others are prefabricated and pre-engineered steel building manufacturers, building components (doors, windows) fabricators, manufacturers of furniture, steel utensils, steel kitchen equipment and other small fabricators manufacturing miscellaneous steel items. Most of these activities are labour intensive. There is a large informal fabrication sector which is unquantifiable, given the nature of their business. Total facilities under this activity are estimated at 150,000 tons per annum.

f) Aluminium industries

There is only one manufacturer (Kalu Works Limited) of aluminium flat/rolled products in the entire Sub-Saharan Africa. Kalu Works Ltd imports aluminium ingots. The company's value chain starts from the melting furnace, cold rolling mill, slitter, blanking line, tension leveler, and color coating line and roll former. The company manufactures aluminium circles and utensils (pots and pans), aluminium plain sheets, tread plates, aluminium roofing sheets, aluminium pre-painted coils, unpainted aluminium coils, aluminium closure stocks.



It has a melting facility of approximately 29,000 MT/month, rolling capacity of 28,000 MT/annum and an aluminium color coating capacity of 16,000 MT/annum. Kalu Works manufactures and sells around 12,000 MT per annum (total of all products) in the domestic market export and currently utilizes only 40 % of its capacity. The company imports around 8000 MT/ of aluminum ingots per annum. Additionally, Kalu buys around 4900 MT/annum of local aluminium scrap, and exports around 25% of total sales.

Others in this sub sector are aluminium fabricators who are involved in the extrusion of aluminium profiles used in the construction industry.

2.3. Facts and figures about steel industry in Kenya

There are five main export products from the steel sector: iron and steel, metal containers, wire products: nails, screws, nuts, and household equipment of base metal and manufactures of base metal (Table 2.1).

Table 2.1: Export of steel products (MT)

	2013	2014	2015	2016	2017	2018	2019
Iron and steel	155,442.00	137,561.00	124,190.00	139,019.00	108,717.00	110,471.60	149,325.20
Metal containers	2,831.00	3,162.00	2,542.00	3,569.00	2,456.00	1,931.70	3,120.30
Wire products: nails screws, nuts, etc	9,232.00	8,265.00	6,585.00	4,581.00	7,212.00	5,121.90	4,083.30
Household equipment of base metal, n.e.s.	10,401.00	9,841.00	7,614.00	9,690.00	6,225.00	3,550.50	3,741.80
Manufactures of base metal, n.e.s.	7,572.00	8,217.00	11,889.00	10,741.00	11,480.00	12,107.60	11,955.70

Data source: Republic of Kenya, 2017 and 2020

The local steel industry is heavily dependent on imported raw materials (Table 2.2), as no local sources have been developed to date. The main raw materials imported include iron and steel, non-ferrous metals and metal structures and their parts. The quantity of imported iron and steel has grown over time, from 1,217,865 MT in 2013 to 1,594,243.50 MT in 2019 or 5.55% (31% growth).

Table 2.2: Import of steel products (MT)

	2013	2014	2015	2016	2017	2018	2019
Iron and Steel	1,217,865	1,196,273	1,530,606	1,443,869	1,374,713	1,313,822	1,594,243.50
Non-ferrous metals	52,588	53,315	51,010	51,397	46,036	46,322	40,226
Structures and parts of structures, n.e.s., of iron, steel or aluminium	52,612	66,405	95,065	103,981	69,821	59,364	54,920.20

Data source: Republic of Kenya, 2017 and 2020

The high dependence of the steel sector on imported raw materials exposes it to foreign exchange risks. As such, a sudden depreciation of the Kenyan currency can have devastating effects on the sector. In addition, this further implies that the sector requires a very efficient logistical supply chain system owing to the weight and bulkiness of the imported raw material.

2.4. Challenges facing the steel industry in Kenya

The below pointers give an overview of key challenges that the steel industry faces in the country:

■ **Global overcapacity:** This can be traced from China which supplies more than 50% of global steel output.

■ **High cost of electricity:** Kenya manufacturers pay high energy which makes them uncompetitive in the export markets. For smelters, power takes up almost 40-50% of the total conversion cost. Power cost alone per MT works out to over US\$100, whereas internationally same amount is the total conversion cost.

■ **High dependence on imported raw materials:** Almost all raw material requirements for local steel plants are imported, primarily because iron ore and coal deposits in Kenya are yet to be exploited. This increases the vulnerability of the sector to external shocks.

■ **Levies:**

- 1.5% import declaration fee (IDF) is an additional cost, which makes Kenyan manufacturers uncompetitive within the regional markets. The 1.5% IDF translates into a 1.5% cost disadvantage.
- Railway development levy (RDL) of 1.5% translates to a 1.5% cost disadvantage to domestic manufacturers.

■ **Poor infrastructure:** The road network within the cities and towns (industrial areas) is in a poor state and requires immediate attention.

■ **High borrowing cost and short repayment period:** Interest costs are high.

■ **High risk on extended credits to customers:** Manufacturers take a high risk due to tight liquidity in the market. There is no suitable cover for this risk. There is a need to develop and enforce a system to protect suppliers against defaulters.

■ **Domestic overcapacity:** In the past few years, there have been many new entrants both local and overseas, which have added to the existing over-capacity in the country.

■ **Duty free imports on foreign funded government projects:** Such projects have only 40% as local content.

■ Importation of undervalued and substandard goods.

■ Issues pertaining to VAT regulations with regards to exports.

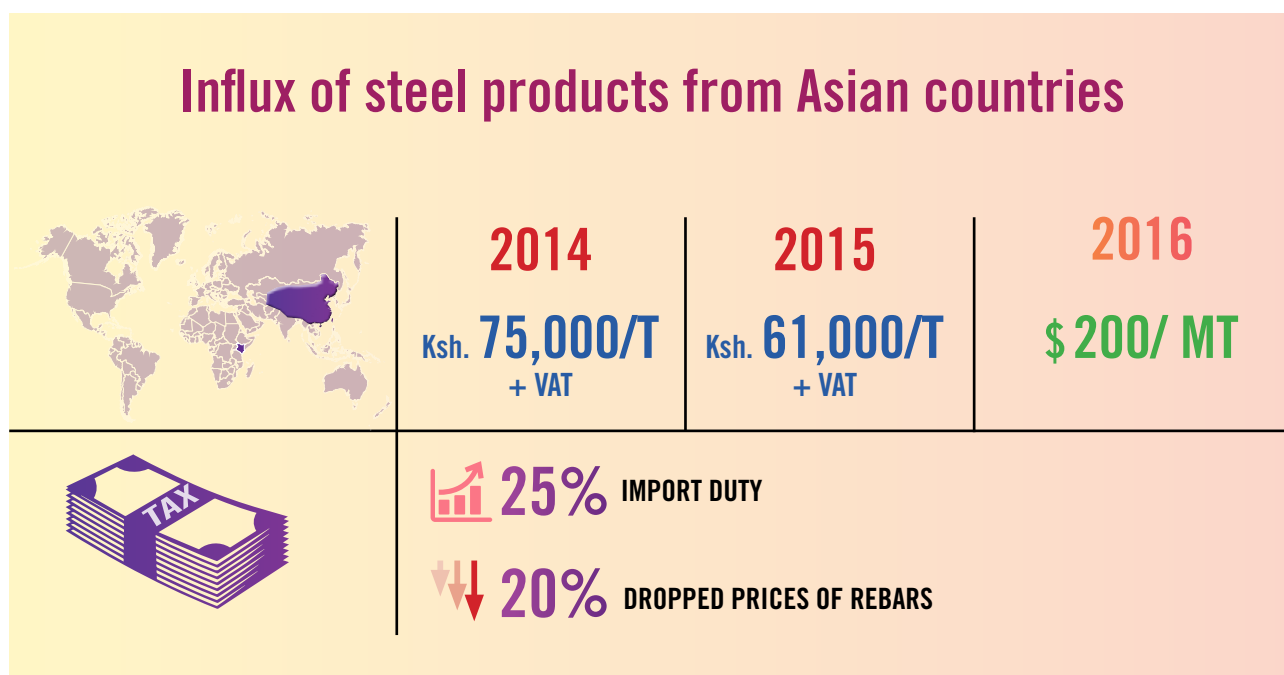
2.5. Steel Industry before and after imposition of enhanced import duties

A few years ago, the steel sector was at its deathbed due to influx of steel products from Asian countries. In January 2014, the average price of rebars in Kenya was Ksh. 75,000 per tonne, plus VAT. In August 2015, the average price of rebars was Ksh. 61,000 per tonne plus VAT. Despite the imposition of 25% import duty on imported rebars and finished steel manufactured in Kenya, the price of rebars dropped by almost 20%. The duty failed to protect local manufacturers and companies, who were expected to lower their prices even further if they were to compete with Chinese steel.

The situation was further compounded by the high cost of inputs in Kenya. Additionally, if non-tariff costs are compared with other countries (including East African partner states), the challenges facing Kenya's steel sector include imports for Government projects funded by other countries, the poor quality of power including voltage surges and frequent failures, a lack of highly skilled personnel, the volatile pricing and availability of scrap metal, the inadequate control of scrap metal trade by the Government and the imposition of duty by Customs on imports of spare parts.

In recent years, the steel industry in Kenya has gone through turbulent times due to a drop in international prices of steel. This was due to oversupply of steel in the world market owing to a slowdown of the economy in most parts of the world. With dumping of finished steel in the Kenyan market in the past, many steel factories were forced to close down their operations and/or downsize manufacturing and import finished steel. Import surges of steel from Common Market for Eastern and Southern Africa (COMESA) member states, particularly Egypt that has lower costs of production and subsidies, has significantly eroded available market for locally manufactured steel.

In the past, government increased the import duty on finished steel from 10% to 25% but the industry remained vulnerable from cheaper import dumping. Given this challenge faced by the steel sector, the Government imposed a minimum duty of US Dollars 200 per metric tonne or 25%, whichever is higher from June 2016, giving the sector a much-needed lifeline. This has seen many companies which had closed down their manufacturing plants to restart.

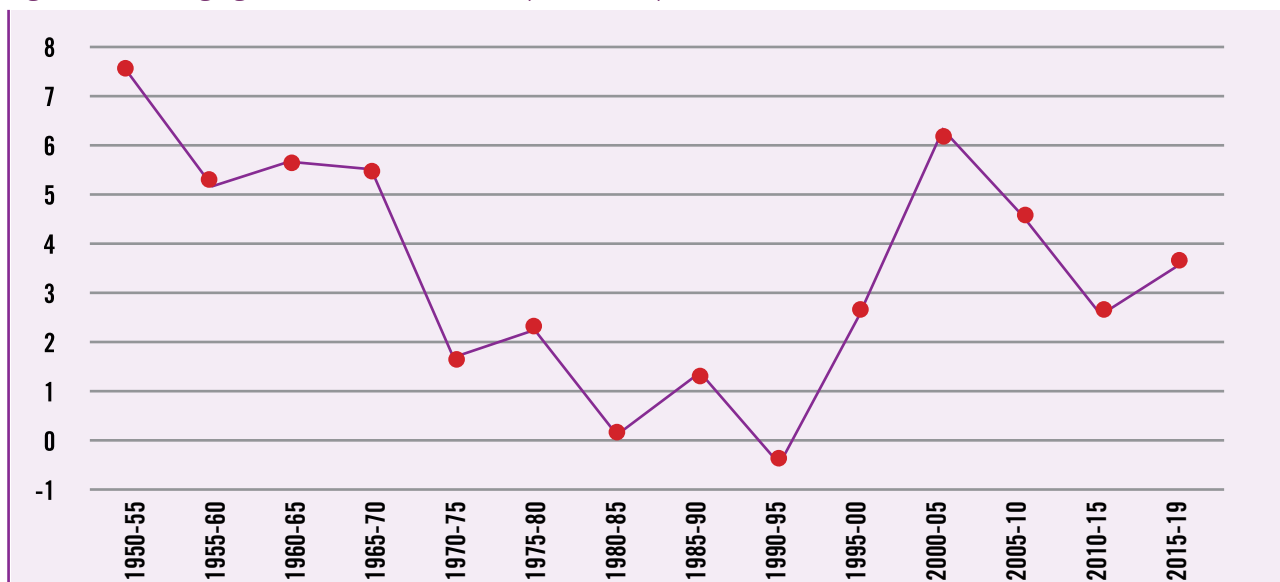


3. Global and regional trends in the steel industry and associated challenges

3.1. Facts and figures of the global steel industry

The growth rate of crude steel production has been on a downward trend for almost half a century, from 1950-1995. There was a sharp growth rate from 1995-2005, followed by a sharp decline till 2015. After 2015, the steel industry in the world has been on a recovery path (Figure 3.1). The steel industry is sensitive to the general performance of the economy. The 1990-2005 period coincides with the period in which economies of the emerging markets recorded high sustained economic growth rates.

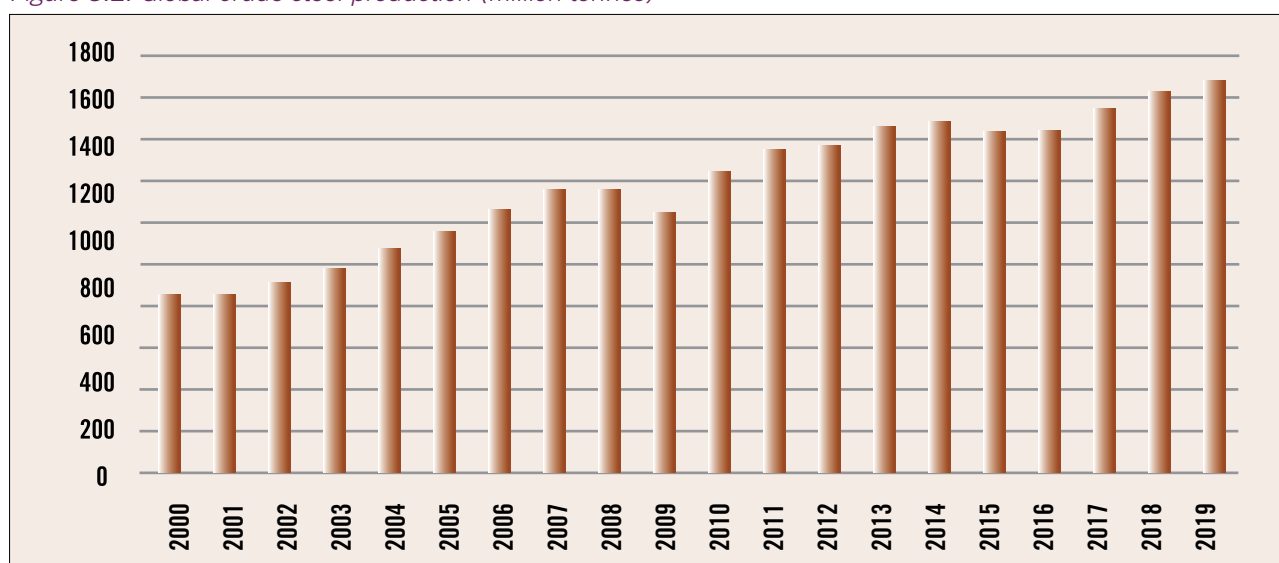
Figure 3.1: Average growth rate of crude steel production per annum (%)



Data source: World Steel Association, 2020a

Since 2000, global crude steel production has been on an upward trajectory (Figure 3.2).

Figure 3.2: Global crude steel production (million tonnes)



Data source: World Steel Association, 2020a

China is the leading producer of steel in the world accounting for 53.3% of global steel production in 2019 (Table 3.1). Global steel production is concentrated among few producers, with 10 countries accounting for 84% of global steel produced in 2019.

Table 3.1: Ten leading steel producing countries in the world (million tonnes)

Country	2019		
	Rank	Tonnage	% share
China	1	996.3	53.3
India	2	111.2	6.0
Japan	3	99.3	5.3
United States	4	87.8	4.7
Russia	6	71.9	3.8
South Korea	5	71.4	3.8
Germany	7	39.7	2.1
Turkey	8	33.7	1.8
Brazil	9	32.2	1.7
Iran	10	25.6	1.4
Sub-total		1569.1	84.0
Global production		1868.8	-

Data source: World Steel Association, 2020a

China which is the leading producer of crude steel (Table 3.1) is also the leading exporter of semi-finished and finished steel products (Table 3.2). In Africa, only Egypt and South Africa are major exporters of steel products. Manufacturers of steel products in Kenya have raised concerns about imports from Egypt which is a Member of COMESA and imports attract 0% import duty. Competition is set to increase in the steel sector from South Africa, when trading under the African Continental Free Trade Area (AfCFTA) commences. Under the East African Community (EAC) Common External Tariff (CET), imports from South Africa attract import duty.

Table 3.2: Exports of semi-finished and finished steel products (MT, million) for some selected countries

Country	2014	2015	2016	2017	2018	2019
Belgium	14,161	15,202	16,724	18,060	17,971	16,936
Germany	24,757	25,147	25,521	26,365	25,999	24,061
Italy	17,328	16,475	17,895	18,190	18,180	17,948
United States	11,961	10,000	9,247	10,211	8,623	7,271
China	92,907	111,556	108,066	74,808	68,767	63,744
India	10,379	7,563	10,325	16,335	11,101	13,356
Egypt	1,153	453	595	1,315	1,453	1,162
South Africa	2,339	2,190	2,194	2,562	2,857	4,302
Tanzania	1	-	-	-	1	-
Kenya	-	1	-	-	-	-
World	457,402	467,442	476,579	462,731	457,583	439,424

Data source: World Steel Association, 2020b

The United States is the leading importer of steel products in the world, followed by Germany and Italy (Table 3.3).

Table 3.3: Imports of semi-finished and finished steel products (MT, million) for some selected countries

Country	2014	2015	2016	2017	2018	2019
Belgium	10,743	12,073	13,007	14,073	14,845	13,006
Germany	24,263	24,817	26,620	27,094	26,584	23,096
France	13,418	13,656	14,570	15,073	14,882	14,589
Italy	16,632	19,936	19,616	20,091	20,619	20,652
United States	41,369	36,485	30,913	35,366	31,726	27,116
China	14,903	13,178	13,581	13,909	14,366	15,511
India	9,477	13,284	9,904	8,882	9,026	8,921
Egypt	6,167	7,881	9,197	5,538	421	592
South Africa	1,418	1,699	1,392	1,186	1,024	1,110
Tanzania	602	667	551	535	732	946
Kenya	1,333	1,708	1,491	1,364	1,415	3,124
World	442,674	452,933	463,342	451,600	444,350	438,457

Data source: World Steel Association, 2020b

3.2. Challenges facing the global steel industry

a) Global excess capacity

Global steel production capacity has more than doubled since the early 2000s due to the rapid growth of construction and manufacturing activities and expansion of infrastructure, particularly in emerging economies (de Carvalho and Sekiguchi, 2015). The implication has been over-supply, low prices, weak profitability, bankruptcies and loss of jobs (de Carvalho and Sekiguchi, 2015). According to Manser (2013), there are five reasons that can explain global overcapacity in steel production:

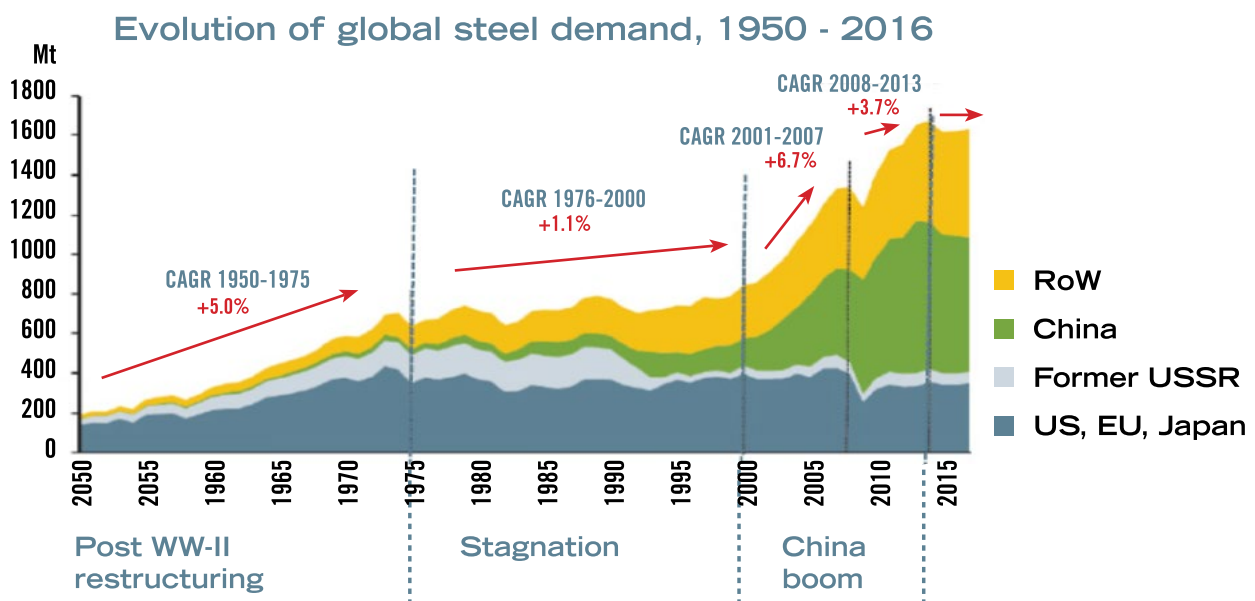
- Technological upgrades have been accompanied by capacity increases.
- Major global producers of steel such as China, India and Russia have high internal transport costs, implying that their domestic markets can suffice for the steel mills.
- Domestic availability of raw materials in Russia, India, Ukraine and Turkey ensures that they are low-cost producers of steel.
- Availability of cheap capital for investment in countries such as China.
- Open international markets avail opportunities to offload excess steel that is produced at low prices.

In the foreseeable future, the imbalance between production and consumption will continue to pose a challenge to both manufacturers and governments around the world.

b) Slow in growth in demand

There is a strong correlation between the use of metal and per capita GDP, though it levels off at higher GDP levels (UNEP, 2013). This is illustrated in Figure 3.3 where the compound annual growth rate (CAGR) of crude steel demand was in tandem with the growth of countries and regions of the world.

Figure 3.3: Evolution of global steel demand, 1950-2016



Source: World Steel Association, 2017c

Demand for steel is expected to be subdued in the medium-term due slower growth in China and slower growing and aging population particularly in Japan and some European countries. Economic growth prospects will also be greatly undermined by the global outbreak of coronavirus, that continues to ravage major economies of the world such as China, Italy and Spain. This will impact negatively on global demand for steel including economies of the emerging markets. According to a United Nations (2019) report, the world's population is projected to increase by 2 billion persons so as to reach 9.7 billion in 2050 from the current 7.7 billion. The population of Sub-Saharan Africa (SSA) is expected to double by 2050 (99% growth). This is expected to lead to rapid urbanization and create the need for buildings and supporting infrastructure. Increased use of steel will reduce pressure on natural resources and in effect, reduce CO2 emissions and therefore support sustainability.

c) High raw material/energy costs

The main reason can be traced from the fact that the main energy source for a blast furnace is coking coal/coke.

d) Solid waste management

A lot of solid waste is produced during the steel manufacturing process. However, this waste contains valuable products that can be reused.

e) Price volatility of raw materials

Price volatility of raw materials is a major concern for manufacturers as it creates uncertainty about future prices and as a result, makes it difficult to plan. For instance, aluminum has seen price movements over the last 10 years from \$1,300 to \$2,700 per metric ton.¹¹ Some of the measures that manufacturers deploy in containing negative effects of price volatility includes the following:¹²

- **Doing nothing:** Manufacturers allow cash flow and profit and loss account to fluctuate in tandem with commodity price movements. This is a common practice by SMEs.
- **Fixed-price supply contracts:** This provides certainty, but a manufacturer may not benefit from a price reduction.
- **Hedging:** Futures, option, forward and swaps for those manufacturers who can afford.
- **Balance sheet:** Involves setting aside cash to absorb price fluctuations internally.

f) Supply-chain vulnerability¹³

Supply-chain vulnerability of the steel industry arises from two reasons:

- The steel industry has a concentrated supply structure, and this increases their vulnerability to disruptions caused by adverse weather conditions and accidents.
- Iron and metallurgical coal quality has been on a decline since 2000s. This has created pressure on the efficiency and environmental performance of the global steel industry.

3.3. Impact of COVID-19 on the global steel industry

The outbreak of the COVID-19 pandemic in late 2019 triggered unanticipated and unprecedented global health and economic crisis, more severe than the global financial crisis of 2009.¹⁴ Lockdown policy measures adopted by many governments across the globe caused global supply-chain disruptions, leading to shortages and delays in the supply of raw materials in a majority of countries. Steel industry performance is highly correlated with the general performance of an economy. COVID-19 has subdued global and individual country growth prospect machinery was mildly impacted. Before the

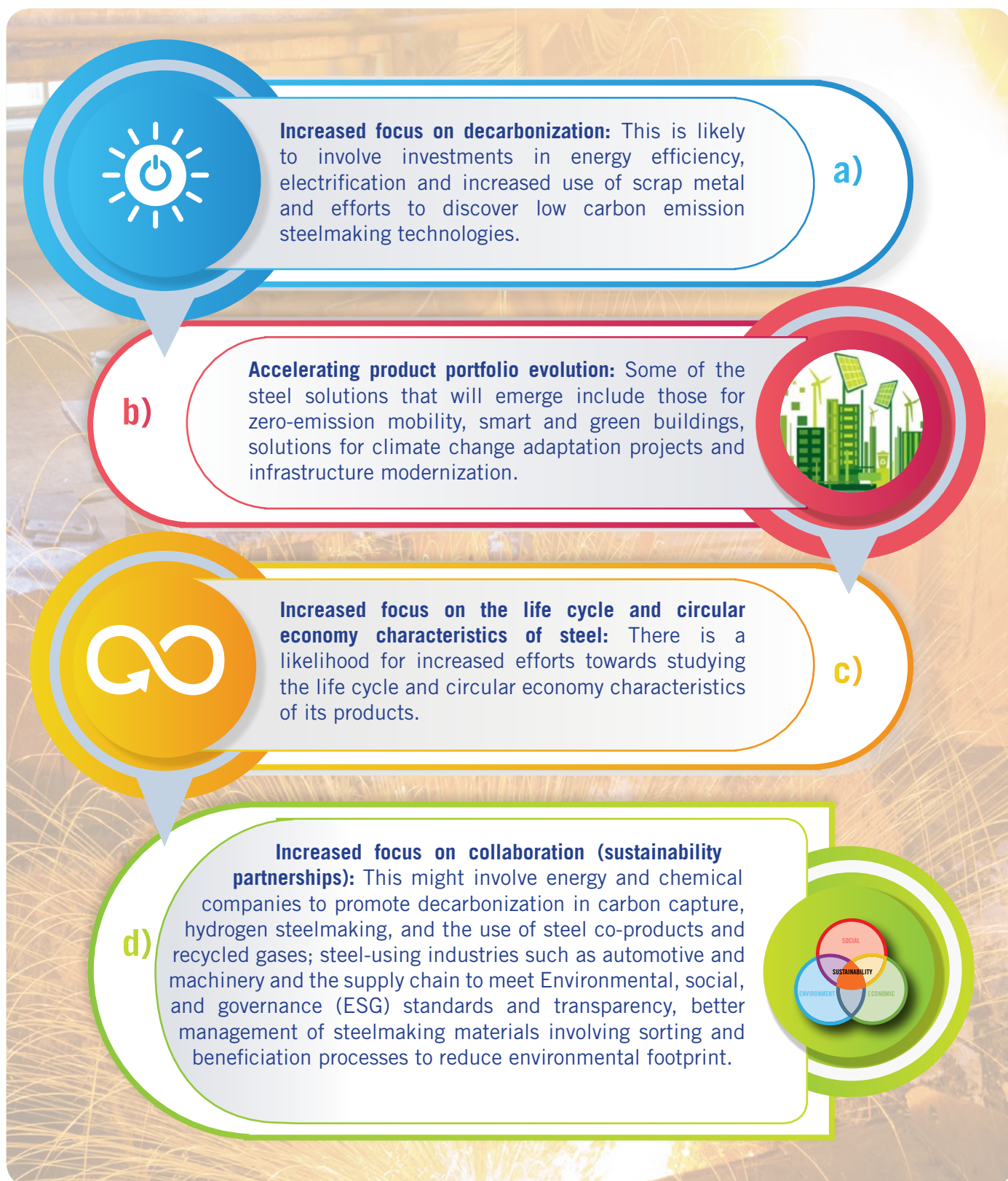
11 <https://www.chai-uk.com/blog-2/2019/09/04-how-metal-price-volatility-is-impacting-manufacturers>

12 <https://www.chai-uk.com/blog-2/2019/09/04-how-metal-price-volatility-is-impacting-manufacturers>

13 <https://www.worldsteel.org/steel-by-topic/raw-materials.html>.

14 https://www.worldsteel.org/en/dam/jcr:514edba3-1429-47da-a3d5-dbf4110402a9/Asia%2520Steel%2520Forum_November%25202020_NHHan.pdf.

outbreak of COVID-19 pandemic, two main transformational forces were technological change and the green transition. The pandemic is likely to accelerate the speed of change.¹⁵ Specific consequences includes the following:



15 https://www.worldsteel.org/en/dam/jcr:c3f93924-e03c-4dfd-933a-c31e0971738c/steelorbis_Dec%25201%25202020_Baris%2520Cifti.pdf.

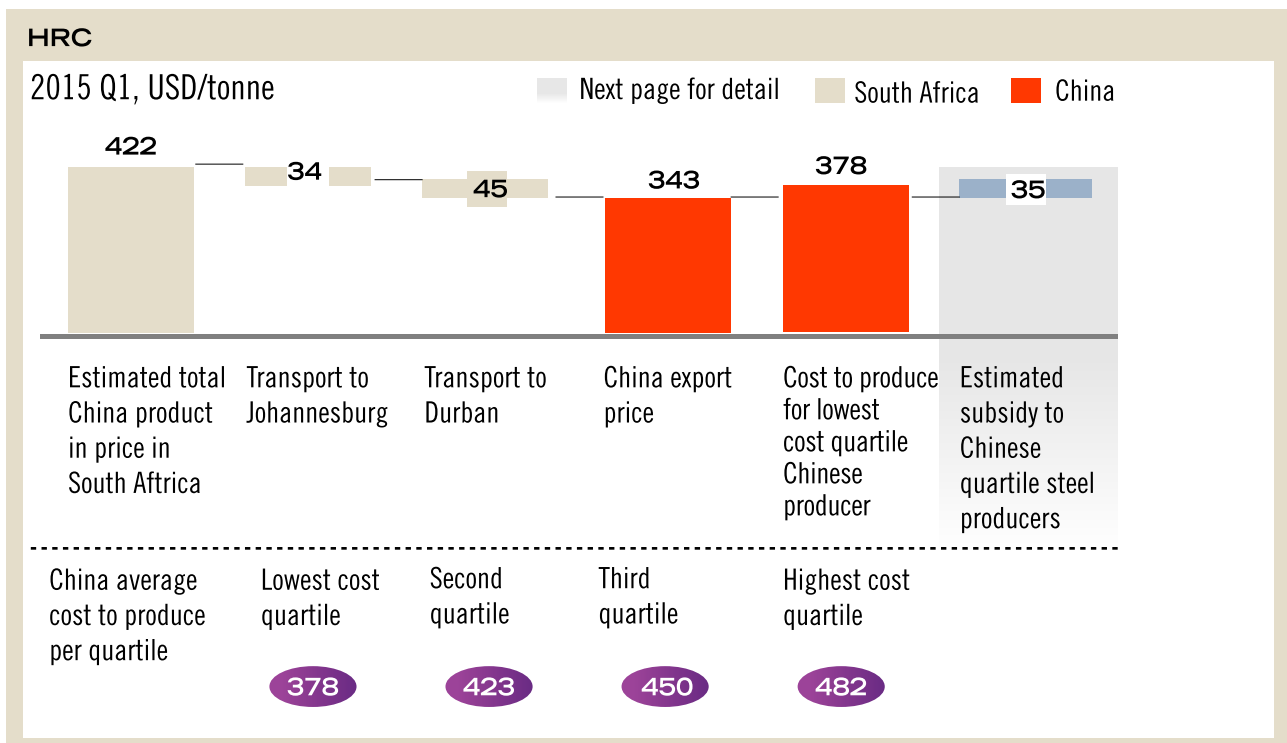
4. International experiences meant to support domestic steel industry



a) China

China produces and consumes about half of the world's steel. In 2016, it was estimated to have an overcapacity of 300 million MT.¹⁶ China has been widely blamed for the Global Steel Crisis of 2015 because it over-built its steel mills capacity in the 2000s.¹⁷ Apart from the overcapacity of the Chinese steel industry, the Government provides direct and indirect support to the manufacturers. In 2015, a Joint Steel Task Team of South Africa estimated that US\$ 122 per ton assistance and subsidies had been provided to steelmakers giving them a strong cost advantage in the global markets. This is the reason why, in 2015, the export price of steel from China was estimated to be US\$ 345, lower than the cost of production of US\$ 375; a clear cost advantage of US\$ 35 (Figure 4.1).

Figure 4.1: China pricing steel production below own cost of production



Source: Joint Steel Task Team of South Africa, 2015

¹⁶ <https://www.reuters.com/article/us-steel-demand-global/global-steel-sector-crisis-fading-demand-set-to-grow-worldsteel-idUSKCN12B1KP>.

¹⁷ <https://www.scmp.com/business/commodities/article/1955015/china-widely-blamed-global-steel-crisis>.

Even though there is scant information on the support offered to steel producers by the Chinese Government, the Joint Task Team of South Africa (2015) estimates that more than US\$ 50 billion was provided to the steel industry through various means including:

- **Government infusion of equity and debt-to-equity swaps:** An example is that the Chinese Government took over the non-performing loan of Shanghai Baosteel and converted it into shares, thus reducing its debt burden.
- **Government sponsored preferential loans and directed credit:** This has been provided to steel producers after satisfying a certain export criterion.
- **An undervalued currency:** The International Monetary Fund (IMF) estimated that the Chinese currency was undervalued by 5-27% with the effect of making Chinese exports cheaper.
- **Land use discounts:** This is accorded to steel producers who set up a manufacturing facility.
- **Majority of steel and aluminium producers are state-owned enterprises (SOEs):** They have access to cheap or free land, free or below-market financing, and forms of public support (such as special regulatory preferences). The SOEs are reported to generate 22% of total industrial profits in China, and they represent 38% of industrial assets.¹⁸
- **Mergers:** This has involved merging private and state-owned steel producers.
- **Preferential tax programs, energy subsidies and reduced valued added tax (VAT):** Support has involved electricity subsidies which have increased by 25% since 2008.

These forms of support from the Government allows Chinese producers of steel and aluminium to undersell producers from market-based economies that must purchase inputs from market-based prices (Haley, 2008).

An increasing number of countries have introduced trade remedy measures to slow down import surges from China. Examples include:

-  **Brazil:** In October 2013, it imposed definitive anti-dumping duties ranging between US\$ 235.59-853.46 per ton on cold-rolled steel sheets from China for a period of five years. This was further extended in October 2019 with import duties ranging between US\$ 175.62-629.44 per ton.¹⁹
-  **Turkey:** In 2017, it imposed 16.9%-22.6% anti-dumping duties on Chinese flat non-rolled steel imports.²⁰
-  **India:** In October 2019, the Government of India proposed to impose anti-dumping duties varying from US\$ 28-200 per ton on flat-rolled steel imports from China for a period of six months.²¹
-  **Colombia:** It imposed an anti-dumping duty of US\$ 419 per ton on imports of wire rod of iron or non-alloy steel or other alloy steel, of circular cross section diameter less than 14 millimeters, with a carbon content less than 0.45% by weight, from China for a period of five years.²²

18 <https://www.epi.org/publication/trump-must-act-now-to-protect-u-s-steel-and-aluminum-administration-delays-have-already-heightened-the-import-crisis-for-tens-of-thousands-of-steel-and-aluminum-industry-workers/>

19 <https://www.globaltradealert.org/intervention/16585/anti-dumping/brazil-extension-of-definitive-antidumping-duty-on-imports-of-cold-rolled-stainless-steel-sheets-from-china-and-chinese-taipei-termination-of-definitive-duty-on-imports-from-finland-germany-the-republic-of-korea-and-vietnam>.

20 <https://www.hurriyetdailynews.com/turkey-imposes-anti-dumping-duties-on-chinese-steel-imports-123256>.

21 <https://in.reuters.com/article/india-steel-imports/india-proposes-anti-dumping-duty-on-chinese-flat-rolled-steel-imports-idINKBN1WU222>.

22 <http://www.irepas.com/?p=2541>.



United States: In September 2019, the US imposed duties of up to 141% on Chinese structural steel.²³



European Union: In 2011, the European Commission imposed duties ranging from 48.3% to 71.9% on Chinese imports of seamless pipes and tubes of stainless steel. This was extended by another five years in March 2018.²⁴

- On September 23, 2011, the International Trade Administration Commission of South Africa initiated an anti-dumping investigation on screw studding of steel and stainless steel imported from China. This was in response to a petition filed by the South African Fasteners Manufacturers Association (SAFMA) on behalf of the SACU industry. The petitioners alleged the existence of dumping on a comparison between the normal value in China and the export price from China. Tariff lines of the products were 7318.15.41 and 7318.15.35. However, on 1st February 2013, the South African authorities terminated the investigation without imposing a duty.²⁵

China has been sensitive to concerns that it is the main cause of global excess steel capacity. For instance, in 2016, the Chinese Government issued “Guidance for the Steel Industry to Reduce Excess Capacity and Resolve Difficulties for Further Development”, planning to reduce 100 to 150 million ton steel production capacity in the next five years with a series of specific measures.²⁶ In 2016, China closed 65 million tons of iron and steel production capacity far beyond a target of 45 million tons.²⁷ The Chinese Government has also taken measures to curb steel exports such as levying export duty on some selected steel products and voluntarily lowering tax rebates for steel products.²⁸ Other measures include advocating for self-discipline, regulating export orders, monitoring of exports to key regions and proactive engagement with concerned Governments, associations and enterprises so as to promote market stability and reduce steel trade frictions.²⁹



b) India

Steel is a highly regarded industry in India and has a full-fledged Ministry-Ministry of Steel.³⁰ India has large deposits of iron ore and is likely to remain self-sufficient for a long time to come (Hyvonen and Langcake, 2012). The Indian steel industry has faced increased pressure from imports from China,

Japan and South Korea. Some of the measures undertaken by the Indian Government to protect its steel industry include:³¹

- Increasing import duties.
- Levying of safeguard duty.
- Making Bureau of Indian Standards (BIS) norms mandatory on steel products.
- Imposition of Minimum Import Price (MIP) on some selected steel products.³²

23 <https://www.reuters.com/article/us-usa-trade-steel/us-imposes-duties-on-structural-steel-from-china-mexico-idUSKCN1VP2R7>.

24 <http://trade.ec.europa.eu/doclib/press/index.cfm?id=1807>.

25 <https://www.globaltradealert.org/intervention/16144/anti-dumping/south-africa-initiation-and-subsequent-termination-of-antidumping-investigation-on-imports-of-screw-studding-of-steel-and-stainless-steel-from-china>.

26 <http://www.chinamission.be/eng/zgggffz/chneco/t1366481.htm>.

27 <https://www.piie.com/blogs/china-economic-watch/chinas-excess-capacity-steel-fresh-look>.

28 <https://www.piie.com/blogs/china-economic-watch/chinas-excess-capacity-steel-fresh-look>.

29 <http://www.chinamission.be/eng/zgggffz/chneco/t1366481.htm>.

30 <https://www.india.gov.in/my-government/whos-who/council-ministers>.

31 <https://www.steel-360.com/stories/steel/can-mip-protect-the-indian-steel-industry>.

32 MIP can be defined as the equivalent to the weighted average global price of a product and price before injury was caused, whichever is lower.

Table 4.1 provides details of 14 steel categories and corresponding MIP as proposed by the Ministry of Steel in December 2015.

Table 4.1: Products under 14 categories for base and special grades

	Category	Base grade	Special grade
1.	Metallics	250	NA
2.	Semi-finished products	400	460
3.	Hot rolled products	450	510
4.	Hot rolled universal/quarto plates	505	565
5.	Cold rolled	555	615
6.	Coated flat steel	660	720
7.	Colour coated, pre-painted	760	NA
8.	Cold rolled non-grain oriented	915	NA
9.	Steel pipes and tubes	515	580
10.	Seamless tubes and pipes	775	925
11.	Stainless tubes and pipes	2,417	3,404
12.	Stainless steel ingots, other semi-finished finished products	2,117	3,104
13.	Flat rolled products	1,772-2,217	3204-3435
14.	Bars, rods and sections of stainless steel	1,871	2,759

Source: Steel 360 Magazine (Feb 16 – Issue), 2016



c) United States

The United States; steel industry is at the core of its competitiveness. Its steel industry is the backbone to the automotive sector, bridges, buildings, railways, machinery and the national military industry. There are three main drivers of domestic demand of steel in the United States:³³

- **Construction:** This is primarily through public expenditure on infrastructure including bridges and tunnels and private sector investment in the construction of commercial and residential properties. The real estate sector was deeply affected by the global financial crisis of 2008-2009 and it is yet to recover fully.
- **Automotive:** Car makers in the US offer a sizeable market to steel makers. However, car makers have resulted to cheap and lighter materials, impacting negatively on steelmakers' profitability.
- **Machinery:** Manufacturers of appliances, military equipment and other machinery have depended on domestic steel production. The manufacturers of machinery are seeking to use less expensive polymers to reduce their cost of products and this is likely to impact negatively on the steelmakers.

33 <https://www.marketresearch.com/SBI-v775/Steel-Outlook-7114286/>.

The US steel and aluminum industry has faced severe pressure from excess production capacity and unfair trade practices from China, South Korea, Russia, India, Japan, Taiwan, Turkey, Brazil and Vietnam (Scott, 2018).³⁴ This is what prompted President Trump to impose 25% duties on imported steel articles and 10% duty on imported aluminum articles to protect US national security in March 2018.³⁵ However, these measures proved ineffective because of a significant increase in imports of products made of steel and aluminum. Thus, in a presidential proclamation dated 24th January 2020, President Trump expanded tariff coverage to include products made of steel and aluminum such as nails, tacks, staples, cables, certain types of wire, and bumpers and other parts for cars and tractors with effect from 8th February 2020.³⁶



d) South Africa

The steel industry in South Africa contributes about 1.5% of GDP and directly employs at least 190,000 people (Joint Steel Task Team, 2015). Since 2010, South Africa's steel industry has been on a constant decline caused by cheap steel imports leading to reduced profitability and low-capacity utilization of steel mills.³⁷ According to the Department of Trade and Industry (DTI) in South Africa (2018), apart from the residual effects of the Global Steel Crisis, other challenges facing South African steel industry includes:

- Inadequate investment in plant maintenance, equipment and upgrades.
- Increased competition from low-priced imported finished goods.
- High logistics costs.
- High cost of electricity and other administered charges.
- Difficulties in accessing fairly priced scrap metals.
- Discontinued production of certain key primary steel products (particularly higher value-added flat products) for mining, tooling, rail and automotive applications.

After the onset of global steel crisis of 2015 which caused oversupply, depressed prices and import surges, the South African Government created a Joint Steel Task Team to explore ways to forestall possible collapse and loss of capacity for the steel industry.³⁸ Some of the short-medium term measures that the Government of South Africa implemented include:

- Increase the general rate of customs duty on primary steel products to 10% and safeguard measures for a period of 3 years on hot rolled coil and plate products
- Downstream support measures such as tariff review of some selected downstream products and the use of rebates in situations where products are not manufactured, or additional value added before export.
- Agreement on pricing of flat products to ensure steel-dependent industries remain competitive while also making sure that upstream steelmakers remain sustainable
- Local procurement by Government.
- Settlement of Competition Commission issues with AMSA.

34 <https://www.epi.org/publication/trump-must-act-now-to-protect-u-s-steel-and-aluminum-administration-delays-have-already-heightened-the-import-crisis-for-tens-of-thousands-of-steel-and-aluminum-industry-workers/>

35 <https://www.nytimes.com/2018/03/01/business/trump-tariffs.html>.

36 <https://www.nytimes.com/2020/01/27/business/economy/trump-steel-tariffs.html>.

37 https://www.thedti.gov.za/parliament/2018/SA_Steel.pdf.

38 <https://www.iol.co.za/business-report/economy/government-intervenes-to-save-steel-sector-7803948>

- Establishment of a Steel Development Fund worth R. 1.5 billion to support downstream sectors.
- Investment support through 12i tax incentives and incubation support for SME development.³⁹
- Development of reference price by South African Revenue Service (SARS) for downstream products to address low priced imports.
- Development of a short term negotiated electricity pricing framework for energy intensive users to be implemented.
- Participating in G20 meeting to address global steel over capacity.



e) Egypt

The Egyptian steel sector relies heavily on reinforcing bars (rebars) that account for around 80% of all steel sales in Egypt. Steel billets are used as raw materials for the production of rebar (Salim, 2006). The main types of steel produced by Egypt comprises of:

- Carbon steel-long and flat steel.
- Stainless steel-long and flat steel.



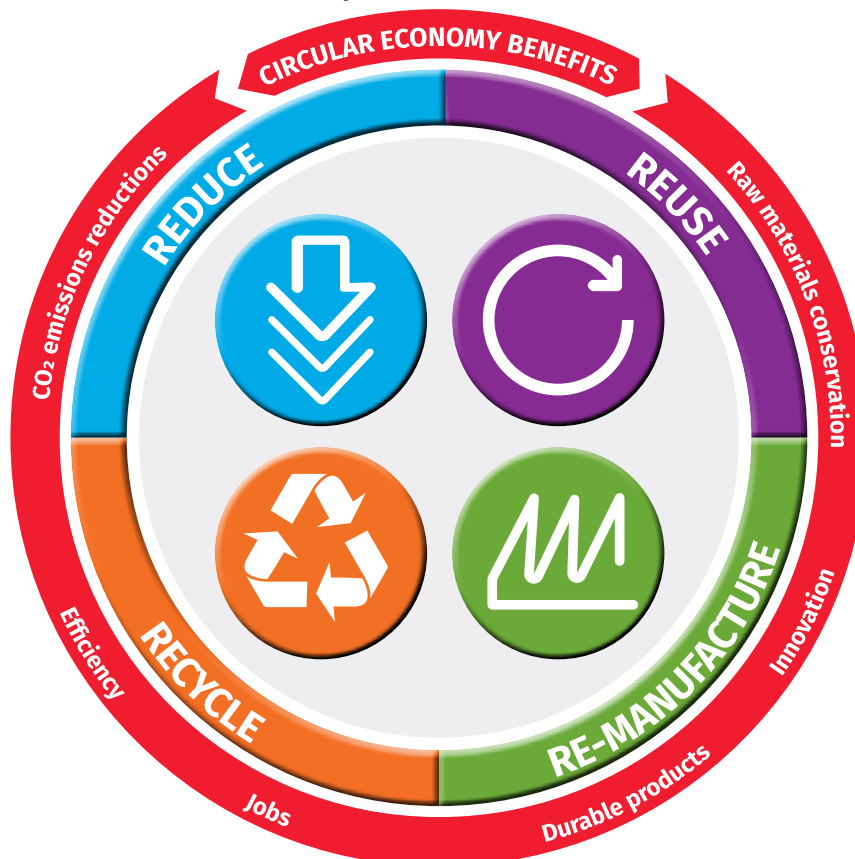
³⁹ The 12i Tax Incentive is designed to support Greenfield investments (i.e., new industrial projects that utilize only new and unused manufacturing assets), as well as Brownfield investments (i.e., expansions or upgrades of existing industrial projects). The incentive offers support for both capital investment and training (see: https://www.thedti.gov.za/financial_assistance/financial_incentive.jsp?id=45&subthemeid=26).

5. Steel production, sustainable development and emerging issues

5.1. Steel production in the circular economy

The circular economy can be referred to as a move from linear business models, in which products are manufactured from raw materials, used and then discarded, to circular business models where products or parts are repaired, re-used, returned and recycled (World Steel Association, 2017a). The concept of circular economy is consistent with that of sustainable development which takes into account the interplay between environmental, social and economic factors.⁴⁰ There are four keywords that define the advantage of steel in a circular economy: reduce, reuse, remanufacture and recycle (Figure 5.1).

Figure 5.1: Benefits of steel in a circular economy



Source: World Steel Association, 2017a

⁴⁰ According to the Brundtland Report of 1987, sustainable development can be defined as the development that meets the needs of the present without compromising the ability of the future generations to meet their own needs.

A sustainable future within the context of the steel industry faces the following challenges:

a) Recycling

Steel has a unique capacity in that it can continuously be recycled without losing its properties or performance (World Steel Association, 2017a). The utilization of steel scrap is crucial in reducing carbon emissions from the steel life cycle. Recycling saves energy, raw materials and landfill space in line with the concept of sustainable development (Javaid, Essadiqi, Bell and Davis, 2006). Recycling increases availability of metal for societal use, so long as there exists potential sources and recycling technologies (UNEP, 2013). There are three categories of steel scrap (Bowler, *et al.*, 2015):

- **Home scrap:** Also referred to as runaround scrap is usually in the form of trimmings or rejects produced within a steel mill during the manufacturing process and can be quickly be reprocessed since it does not leave the factory.
- **New scrap:** Also known as prompt scrap, it is generated by fabricators of steel products and can also quickly be reprocessed.
- **Old scrap:** Also called obsolete scrap, it is steel that is discarded at the end of the product life cycle. It comprises of junk vehicles, old appliances and machinery, old railroad tracks, steel from demolished buildings amongst others. There are also possibilities of recovery of high-tech and high-value metals from different sources such as waste electronic and electrical equipment (WEEE), lamps, LEDs, batteries etc. (UNEP, 2013).

There are several advantages of steel recycling including:

- Raw material conservation where on average, 1400 kg of iron ore, 740 kg coal and 120 kg of limestone are saved.
- 70% energy saving.
- Job creation through scrap collection, separation and recycling.

Availability of steel scrap is usually limited due to long life of steel products such as buildings and bridges. Policy measures can be instituted by placing emphasis on recyclability and design for dismantling (Worldsteel Association, 2017a).

b) Use of co-products

Co-products can be defined as the materials that are produced in parallel to or, as a consequence of the production of a primary product and which also have a potential value (Worldsteel Association, 2018). The use of co-products by substituting resources in other industries can reduce CO₂ emissions (Worldsteel Association, 2017a). The main solid co-products out of iron and crude steel production are slags (90% by mass) and dust and sludge. Table 5.1 gives some examples of common uses of steel industry co-products.

Table 5.1: Some examples of common uses of steel industry co-products

	Co-product name	Use
1.	Blast furnace slag	Substitute for clinker in cement-making
2.	Steelmaking slag	Aggregates in road construction, soil improvement
3.	Process gases	Heat and electricity production
4.	Dust and sludge	Internal and external use of iron oxides and alloying elements
5.	Petrochemicals from coke making	tar, ammonia, phenol, sulphuric acid and naphthalene for the chemical industry
6.	Emulsions from mills and used oil	Reducing agent in blast furnaces or used in coke plants.

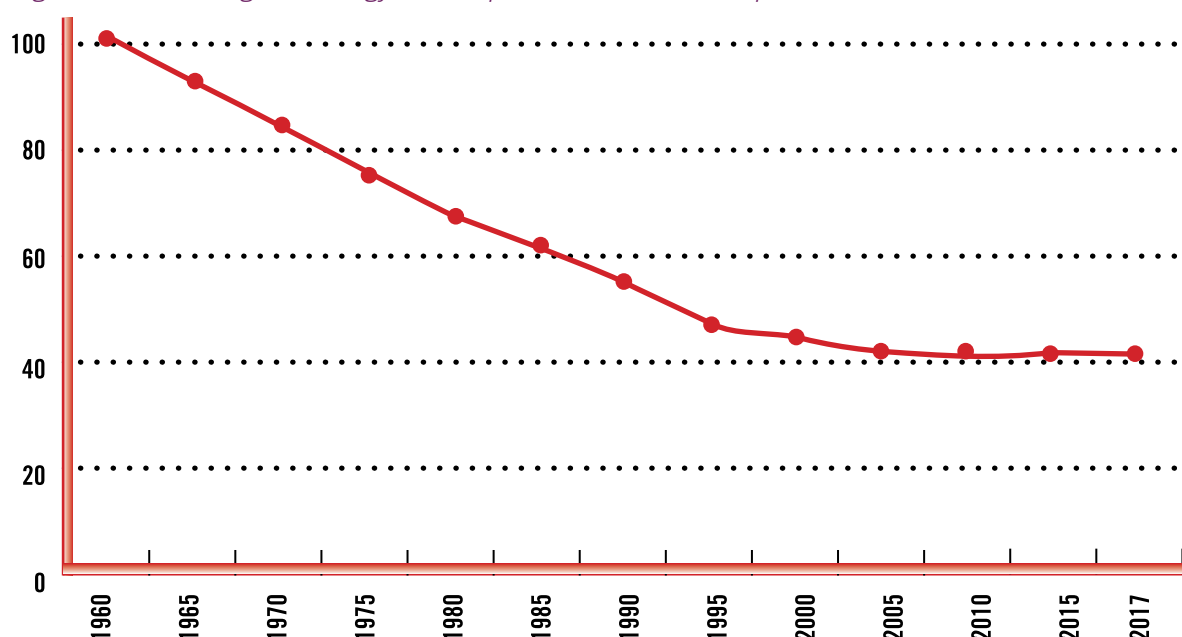
Source: World Steel Association, 2018a

Using a steel industry co-product as a substitute for an equivalent product will improve resource efficiency and contribute to the circular economy (World Steel Association, 2018a).

c) Energy efficiency

According to the World Steel Association (2017a), technological advancements have reduced energy consumption per ton of steel by 61% as shown in Figure 5.2 and due to dramatic improvements in energy efficiency over the years, there is limited scope for further improvements on the basis of existing technology.

Figure 5.2: Indexed global energy consumption/ton of crude steel production



Source: World Steel Association, 2017

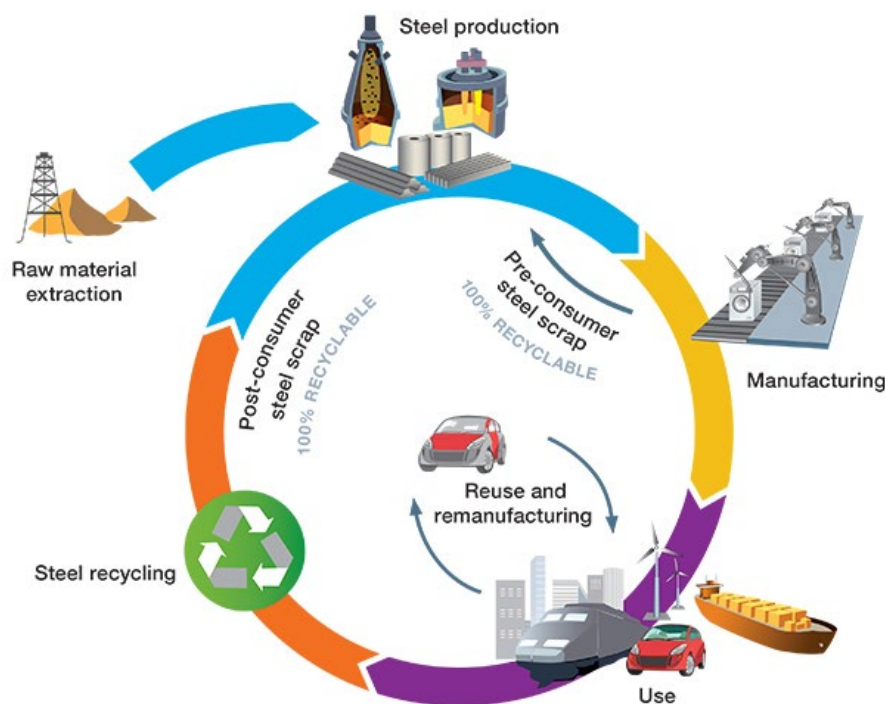
d) Use of finished steel

Steel has a very long service life in many applications and as a result, the contribution of modern steel in improving energy efficiency of buildings, plants, machinery and transportation is very important. It is estimated that, avoided carbon emissions from the use of high-grade applications are on average six times higher than those from the production of the advanced steels (World Steel Association, 2017a).

e) Life cycle approach

Life cycle Assessment (LCA) is a tool that evaluates the environmental, economic and social performance of materials and consumer products and provides a holistic approach that considers the potential impacts from all stages of manufacture, product use and end-of-life (reuse, recycling or disposal) (Worldsteel Association, 2017b). Figure 5.3 shows lifecycle of steel.

Figure 5.3: Life cycle of steel



Source: World Steel Association, 2017b

5.2. Climate change mitigation

In the steel production process, carbon is produced from a chemical reaction involving coke and coal (carbon) and iron ore in a blast furnace (World Steel Association, 2019). Modern steel plants are highly energy efficient and operate close to the limits of thermodynamics. As a result of adopting best practices and technologies, not only has energy efficiency been maximized, but carbon emissions have also been minimized. At the global level, there are projects that are in their early stages of development and others are being piloted to reduce carbon emissions and can be categorized as follows (Table 5.3):

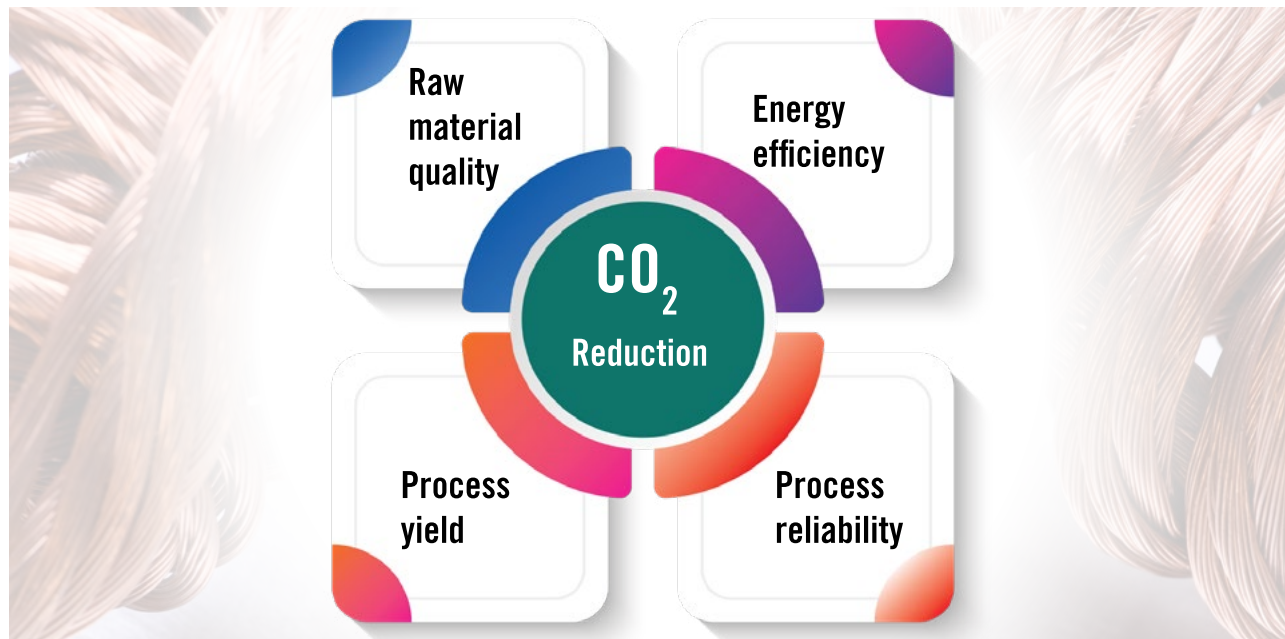
Table 5.3: Global projects to reduce CO₂ production in the steel manufacturing process

Approach	Description
Hydrogen as a reducing agent	Avoids carbon and uses hydrogen to reduce iron ore thereby averting the creation of CO ₂ while producing H ₂ O (water) instead.
Carbon Capture and Storage (CCS)	Generates a clean and concentrated CO ₂ stream that can be captured and stored.
Carbon Capture and Utilisation (CCU)	Uses the components of the co-product gases from existing processes to produce fuels or input material for the chemical industry.
Biomass as a reducing agent	Can partially substitute coal for biomass such as charcoal.
Electrolysis	Reduces iron ore using electricity

Source: World Steel Association, 2019

In 2019, World Steel Association launched a programme called “Step Up” aimed at supporting companies to attain the highest level of operational efficiency. The program covers four areas as shown in Figure 5.4.

Figure 5.4: Four-stage efficiency review process



Source: World Steel Association, 2019



Optimal raw materials selection and use: There is a direct relationship between energy intensity and CO₂ emissions and the quality of iron ore and coking coal. Processes such as beneficiation of ore and coal at the source, use of carbon-lean or hydrogen containing fuels, increased use of scrap can significantly enhance operational performance.

Energy efficiency and minimizing waste: Some of the measures include heat or energy recovery from solid and gas streams, coke dry quenching, cogeneration units, electricity savings (aiming toward self-sufficiency) etc.



Improving yield: Improving yield leads to more output and consequently a reduction in energy and raw material use.

Process reliability: Constant maintenance of steel plants ensures process reliability and in effect reduces losses in quality and process time and thereby reduces energy use in steel processing.



5.3. Safety and health

Safety and health of persons working in the steel industry is critical. There has been notable improvement globally as reflected by reduction in the Lost Time Injury Frequency Rate (LTIFR) where the injury rate per million hours has reduced from 4.55 in 2006 to 0.97 in 2017, a 78.67% reduction (World Steel Association, 2018b). The ultimate target should be “zero” injury at workplaces. Steel Safety Day is observed annually, every 28th April and it is aligned with the International Labour Organization’s (ILO) World Day for Safety and Health at Work. Most countries have their own occupational health and safety legislation, but steelmakers often follow international standards on occupational health and safety management such as OHSAS 18001 and ISO 45001 (Worldsteel Association, 2018b).



5.4. Water management in the steel industry

Water consumption is always a function of water availability in connection with energy efficiency, treatment efficiency, and legal regulatory and safety and health requirements (Thörner, 2018). Table 5.4 shows different water types in the iron and steel industry.

Table 5.4: Water types in iron and steel industry

Application	Media	Process
Direct cooling	Water	Hot rolling, casting, quenching
Indirect cooling	Water	Oven cooling (e.g., blast furnace) Machinery cooling
Various applications	Water	Descaling
	Water	Gas washing (e.g., blast furnace)
	Water	Water-jet vacuum pump operation for vacuum treatment of steel
	Water	Material conditioning (e.g., slag granulation)
Further liquid media	Emulsion/cleaning agent Inorganic acids / water	Cold rolling/degreasing bathes Pickling/rinsing

Source: Thörner, 2018⁴¹



⁴¹ Contact water/direct cooling refers to water with direct contact with (hot) steel e.g., hot rolling, quenching while non-contact water/indirect cooling refers to water without product contact, e.g., machine or oven cooling.

6. Domestic opportunities that can be harnessed to promote the Steel Sector in Kenya

6.1. Developing domestic capacity to supply raw materials

a) Iron ore

Kenya has large iron ore deposits in Kitui, Taita Taveta, Homa Bay and Kakamega Counties. Commercial exploitation of iron and steel deposits will not only allow for the development of an integrated iron and steel plant in Kenya, but also billet casting facility to supply raw materials to steel plants in Kenya and the surplus can be exported to the region.⁴² Exploitation of iron deposits will reduce overreliance of the steel industry in Kenya from imported raw materials mainly from China and India. The COVID-19 crisis has served to illustrate the risks associated with importation of raw materials when global supply chains get disrupted.



b) Coal



There are commercially viable coal deposits in Kitui and Lamu Counties. Coal deposits in Mui Basin, Kitui County are estimated to be valued about US\$ 40 billion.⁴³ A Chinese company Jung'u was awarded a mining concession in 2010 to mine coal in Mui Basin. However, this created a conflict with the local community because at least 100,000 residents would have been relocated from their ancestral land. The genesis of this conflict is that the residents were neither involved in the negotiations for their resettlement nor the compensation to be paid.⁴⁴ This conflict over the Mui Basin coal mining is yet to deescalate. This is because, in 2019, residents petitioned Parliament to stop coal mining due to environmental, livelihood and health concerns.⁴⁵

c) Scrap metals

Steel is the most recycled material in the world and the most sustainable material. This is due to three attributes: infinite recycling without loss of properties; it is a permanent material and easy magnetic separation and



⁴² <http://www.industrialization.go.ke/images/downloads/investment-opportunities-in-manufacturing-sector.pdf>.

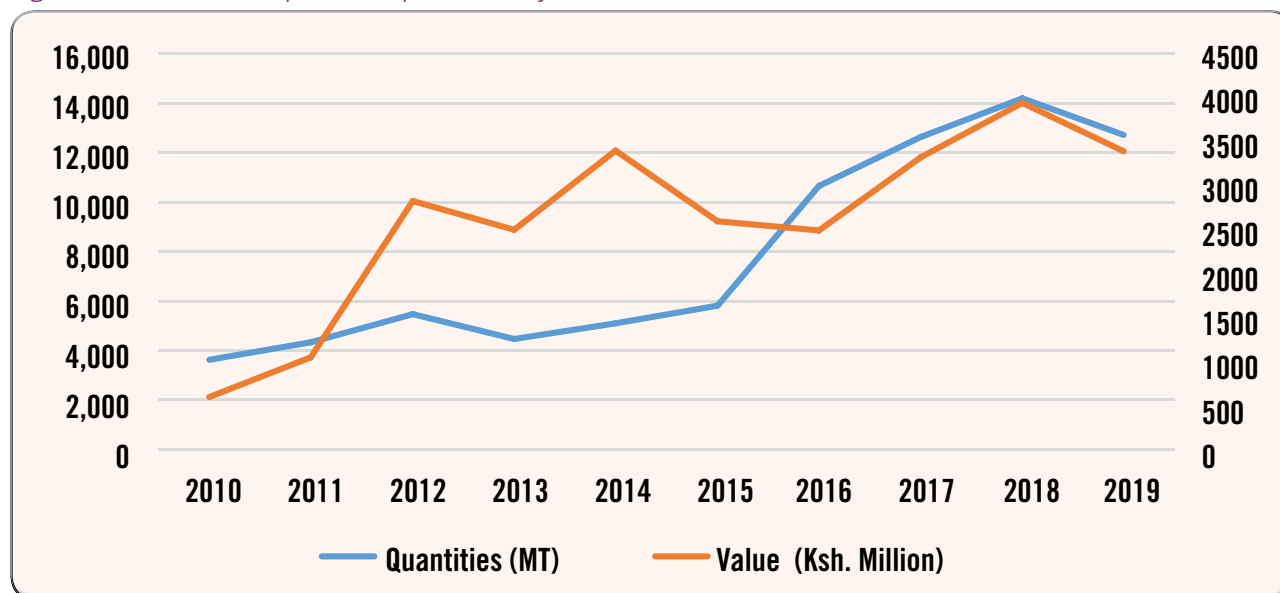
⁴³ <https://www.standardmedia.co.ke/article/2001303915/why-kenya-should-be-wary-of-coal-mining-pitfalls>.

⁴⁴ <https://www.ciel.org/wp-content/uploads/2017/11/Coal-mining-disrupts-peoples-livelihoods-in-Mui-basin-Kenya.pdf>.

⁴⁵ <https://www.business-humanrights.org/en/kenya-locals-petition-parliament-to-stop-coal-mining-due-to-environmental-livelihood-health-concerns>.

recovery.⁴⁶ The Government through the Miscellaneous Fees and Levies Act 2016 introduced a 20% export levy on scrap metal products of different tariffs. However, as can be seen in Figure 6.1, the export levy has not deterred export of scrap. One of the main reasons for introducing the export levy was to make available raw materials for local steel producers. metals. This is because both volume and value of scrap metal exports have been on an upward trajectory since 2016 (Figure 6.1).

Figure 6.1: Value of scrap metal exports in Kenya (Ksh. million)



Data source: Republic of Kenya, 2017 and 2020

The Government, through the Scrap Metal Act 2015 created the Scrap Metal Council in an effort to regulate scrap metal industry in Kenya. The Council was constituted in 2019 but it is yet to become fully operationalized to discharge its mandate.

6.2. Inter-industry industry linkages with the Steel Sector

a) Buildings and infrastructure

Construction industry is the main user of steel products and accounts for more than 50% of world steel demand.⁴⁷ There are also multiple uses of steel in buildings and infrastructure. The most common applications are as follows:

For buildings:⁴⁸

- Structural sections: these provide a strong, stiff frame for the building and make up 25% of the steel use in buildings.
- Reinforcing bars: these add tensile strength and stiffness to concrete and make up 44% of steel use in buildings.
- Sheet products: 31% is in sheet products such as roofing, purlins, internal walls, ceilings, cladding, and insulating panels for exterior walls.
- Non-structural steel: steel is also found in many non-structural applications in buildings, such as heating and cooling equipment and interior ducting.
- Internal fixtures and fittings such as rails, shelving and stairs are also made of steel.

46 <https://www.worldsteel.org/steel-by-topic/sustainability/materiality-assessment/recycling.html>.

47 <https://www.worldsteel.org/steel-by-topic/steel-markets/buildings-and-infrastructure.html>.

48 <https://www.worldsteel.org/steel-by-topic/steel-markets/buildings-and-infrastructure.html>.

For infrastructure:

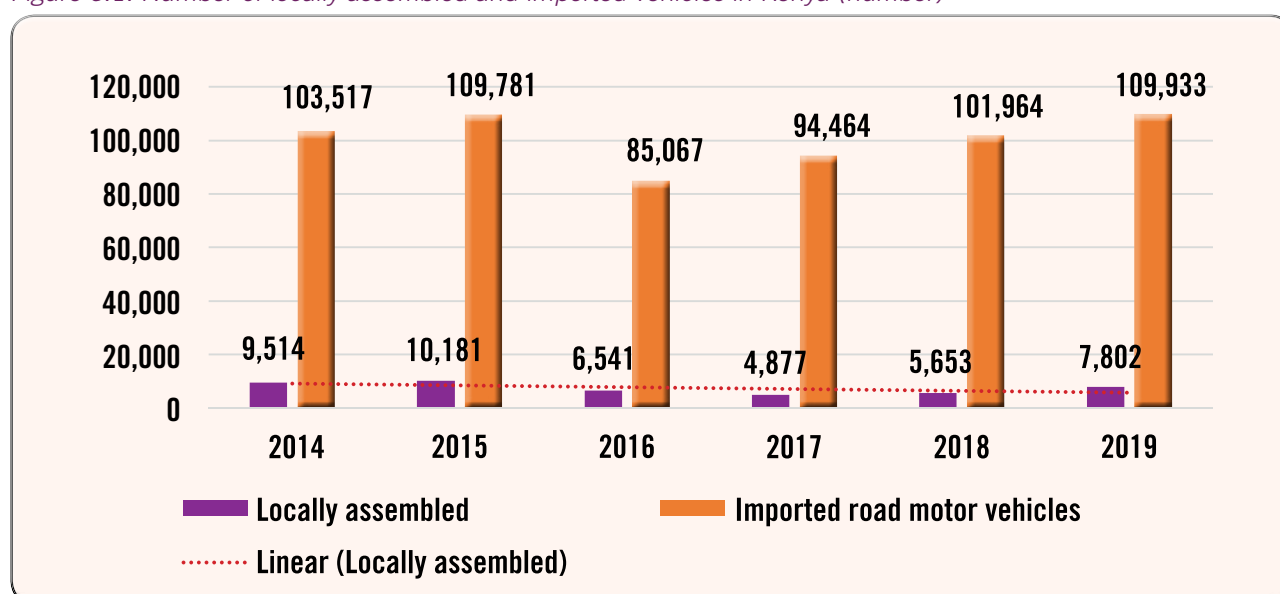
- Transport networks: steel is required for bridges, tunnels, rail track and in constructing buildings such as fueling stations, train stations, ports and airports. About 60% of steel use in this application is as rebar and the rest is sections, plates and rail track.
- Utilities (fuel, water, power): over 50% of the steel used for this application is in underground pipelines to distribute water to and from housing, and to distribute gas. The rest is mainly rebar for power stations and pumping houses.

b) Automotive

The automotive sector is an important source of demand for steel industry in Kenya. World Steel Association estimates that on average, 900kg of steel is used in a car. About 34% is used in the body structure, panels, doors and trunk closures; 23% in the drive train; 12% in the suspension and the remainder on wheels, tyres, fuel tank, steering and braking systems.⁴⁹ Local automotive sector is still very small and rely heavily on imported materials in the form of semi knocked down (SKD) and completely knocked down (CKD) dismantled parts.

Global production of motor vehicles stood at 95.6 and 91.7 million vehicles in 2018 and 2019 respectively; this is according to the International Organization of Motor Vehicle Manufacturers. Kenyans have a heavy preference for imported fully built vehicles as shown in Figure 6.1. This could be because an overwhelming majority of the imported vehicles are used cars and therefore relatively cheaper. The number of locally assembled vehicles declined from a high of 10,181 vehicles in 2015 to 5,653 vehicles in 2018 but increased slightly to 7,802 vehicles in 2019 while Kenya imported 109,933 vehicles in 2019 (Figure 6.1).

Figure 6.1: Number of locally assembled and imported vehicles in Kenya (number)



Data source: Republic of Kenya, 2017 and 2020

Advances in technology and increased preference for lighter vehicles will reduce the amount of steel used per vehicle. The use of advanced high-strength steels (AHSS) is now used by almost every new vehicle design. This is to not only make vehicles lighter, but also optimize vehicle designs with the aim of enhancing safety and fuel efficiency. Therefore, AHSS accounts for about 60% of body structure of a vehicle.⁵⁰

⁴⁹ <https://www.worldsteel.org/steel-by-topic/steel-markets/automotive.html>.

⁵⁰ <https://www.worldsteel.org/steel-by-topic/steel-markets/automotive.html>.

c) Transport

Including automobiles, 16% of world steel is used to meet societal transport needs. Some of the major applications include the making of ships and shipping containers, trains and rail cars, airplanes and infrastructure to make bridges, tunnels, rail tracks etc. Some of the benefits of using steel in transport applications includes-recycled steel and is endlessly recyclable, is strong, safe, durable, cost efficient, remanufacture-friendly and reusable rail tracks.⁵¹

d) Energy

Energy is a critical element of development and it is intrinsically intertwined with steel. This is because, steel is used to supply energy from different sources including fossil fuels, nuclear and renewable sources such as wind, solar or geothermal.⁵² Benefits of using steel in energy production and distribution includes:⁵³

- Contains recycled steel and is endlessly recyclable.
- Is safe, provides structural integrity.
- Durable.
- Strong.
- Cost efficient compared to other materials.
- Precise.
- Steel pipes have lower environmental impact than transporting water or fuels by truck or ship.
- Wind towers are easy to set up and maintain.

e) Other uses

They include packaging, tools and machinery making.

51 <https://www.worldsteel.org/steel-by-topic/steel-markets/automotive.html>.

52 <https://www.worldsteel.org/steel-by-topic/steel-markets/energy.html>.

53 <https://www.worldsteel.org/steel-by-topic/steel-markets/energy.html>.

7. Conclusion and policy implications

7.1. Conclusion

- The study has established that the steel sector is very sensitive to prevailing economic situations in both the domestic and global markets. The steel sector flourishes during economic booms and struggles in periods of recession. Performance of the steel sector is therefore a barometer that can be used to gauge the overall health of an economy.
- It is clear that it is difficult if not impossible for the sector to withstand global turbulence characterized by over-supply of steel, particularly from China. Countries have resolved to imposing anti-dumping duties on some selected steel articles from China, in an effort to cope with import surges and protect domestic steel industry.
- Trade remedies measures, particularly increased duties, appear to be a preferred policy response by countries such as the US, Brazil, India, Colombia and South Africa striving to protect their domestic steel mills from possible closures or reduced capacity utilization.

7.2. Policy implications

Evidence has shown that it is a common practice for countries to impose higher import duties and introduce specific duties as a remedy for unfair trade practices such as export rebates. Kenya should therefore not hesitate to do the same for the domestic steel industry. This is by granting the sector's requests through the annual budget making process.

Export of metal scrap has continued to rise despite the imposition of an export levy of 20%. To further enhance this measure, the study recommends the export levy to be increased to 35% and an equivalent specific duty. This will go a long way in reducing dependence on imported raw materials and enhance job creation in the domestic economy. The Government should also consider imposing a total ban on export of scrap metal as this will not only secure raw materials for the Kenyan steelmakers but also reduce vandalism of public utilities such as power transmission systems and railways among others.

The Cabinet Secretary in charge of Industry should fully operationalize the Scrap Metal Council to discharge its mandate, particularly with a view to ensuring availability of metal scrap for local steel manufacturers.

In order to further reduce overreliance on imported raw materials, the Government should fast track exploitation of iron ore and coal deposits in the country. This will reduce the cost of production and in effect enhance the competitiveness of the steel sector in Kenya.

The government should create a steel fund to support manufacturers as the industry is highly capital intensive. The financial burden required to achieve technological breakthroughs and upgrades should be shared between the Governments and the private sector investors. South Africa created Steel Development Fund to support steel sector in the country. Affordable credit will be critical for the steel industry to be in sync with global practices associated with technological advances and green transition.

Make Kenya Bureau of Standards (KEBS) norms for local steel products mandatory on all imported steel products.

Individual steel companies need to devise their own internal ways of adapting to the changing market conditions characterized by global over-supply of steel products.

The special economic zones (SEZs) are increasingly becoming a favoured policy strategy by the Kenyan government to foster industrialization, Government to consider creating an SEZ specifically for the Metal sector, preferably at the coast or near electricity sources.

The Government should offer a preferential electricity to all steel plants to make them remain competitive.

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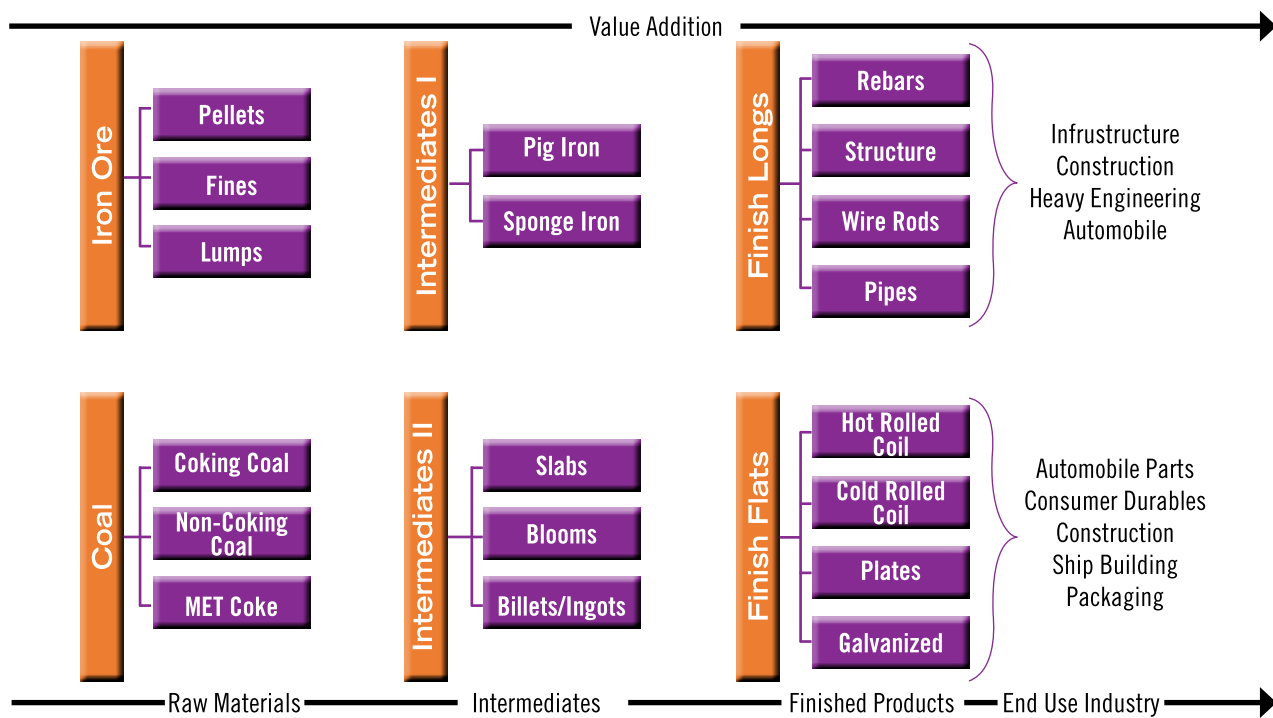
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Annex 1: Steel industry value chain



Source: RBSA Advisors, 2018



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